ALICE L WALTON SCHOOL OF MEDICINE

Drugs to Treat *Plasmodium spp., Babesia spp., and Brucella spp.*

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Adapted from: Drugs to Treat Protozoal and Helminthic Infections

Learning Objectives (3) | Versions

After completing this brick, you will be able to:

1. Justify the use of antimicrobial agents in treating *Plasmodium spp., Babesia spp., and Brucella spp.* based on their mechanism of action.
2. Recognize adverse drug reactions for medications used for *Plasmodium spp., Babesia spp., and Brucella spp.*
3. Design an appropriate evidence-based treatment plan for *Plasmodium spp., Babesia spp., and Brucella spp.*

CASE CONNECTION

BD, a 27 year-old male, is preparing to go on a humanitarian trip to Kenya, Africa to help build a new schoolhouse. He knows he needs to take medications to prevent contracting malaria. He is in your clinic today asking for a prescription for a medication that he can take for prevention.

Knowing that BD is heading to Kenya, an area of East Africa with known resistance patterns to common chloroquine medications, which medication would you prescribe? How soon before his trip, and how long after his trip, would you recommend he take the medication? Consider your answer as you read, and we'll revisit BD at the end of the brick.

GO TO CONCLUSION ↓

As you have learned from previous bricks, malaria (i.e., *Plasmodium spp.*) is endemic in many parts of the world. This makes treatment of malaria important, but prevention for those traveling to endemic areas is paramount as well. The prognosis for malaria is good with appropriate treatment—the parasite is usually cleared from the body within 48 hours. Since the medications used for prevention and treatment are slightly different, we will break up them up by "prophylaxis" and "treatment" below.

Malaria (i.e., *Plasmodium spp.*) Prophylaxis

How do drugs for malaria prophylaxis work? (LO#1,3)

Prophylaxis is impor There are many non-pharmacologic options for malaria prophylaxis that include mosquito netting and insecticides, however, chemoprophylaxis is recommended and has shown a decrease in malaria infections. The most common drug choices include: atovaquone-proguanil, chloroquine/ hydroxychloroquine, doxycycline, mefloquine, and primaquine.

Choosing the most appropriate drug for prophylaxis will include the consideration of the patient's travel itinerary (i.e., are they traveling to a chloroquine resistant area?) and medical history, as well as any contraindications, adverse effects, drug interactions, availability, and cost.

Chloroquine and **Hydroxychloroquine**. We most often refer to "chloroquine" when talking about malaria treatment, however, "hydroxychloroquine" is more often prescribed given the greater availability in the US. Both drugs work the exact same (as described below), but hydroxychloroquine is a derivative of chloroquine, with a hydroxyl group added to its structure, making it potentially less toxic than chloroquine.

Recall from the *Plasmodium* life cycle that in the third stage they enter RBCs and feed on hemoglobin. In order for it feed and grow it must convert the hemoglobin to hemozoin (heme is toxic to the *Plasmodium spp.)*. Chloroquine interferes with digestive vacuole function by increasing the pH and interfering with lysosomal degradation of hemoglobin. The accumulation of toxic heme then damages the parasite's membranes, ultimately leading to its death.

CLINICAL CORRELATION

Hydroxychloroquine often be used in the treatment of some a arthritis. As we mentioned above, this drug increases pH within intracellular vacuoles and alters processes such as protein degradation in the lysosome. They can also affect assembly of macromolecules in the endosomes, and posttranslation modification of proteins in the Golgi apparatus.

It is proposed that the antirheumatic properties of these compounds results from their interference with "antigen processing" in macrophages and other antigen-presenting cells. Acidic cytoplasmic compartments are required for the antigenic protein to be digested and for the peptides to assemble with the alpha and beta chains of MHC class II proteins. As a result, antimalarials diminish the formation of peptide-MHC protein complexes required to stimulate CD4+ T cells and result in down-regulation of the immune response against autoantigenic peptides. This mechanism of hydroxychloroquine is quite unique among other autoimmune medications.

Atovaquone-proguanil. This combination medication is often a first choice for many travelers. Take a look at the Anti-malarial comparison table. Atovaquone-proguanil takes less time to become effective than chloroquine or hydroxychloroquine, which provides a shorter amount of time that the patient has to take the medication, in addition to being effective. This makes it a convenient first option.

Atovaquone acts synergistically with proguanil. Atovaquone selectively inhibits mitochondrial electron transport of the parasite, while proguanil's active metabolite interferes with parasite folate synthesis in DNA synthesis. Together, atovaquone and proguanil affect the erythrocytic and exoerythrocytic stages of development. Meaning, this medication is effective in treating the active liver schizonts as well as t

What is the mechanism of atovaquone/ proguanil?

Doxycycline was discussed previously in the Foundations of Microbiology course. Recall that it is a protein synthesis inhibitor at the 30s subunit. This prevents the *Plasmodium spp.* from completing RNA synthesis and replicating.

Mefloquine is another effective drug for malaria prophylaxis. Its chemical structure is similar to quinine, which is common internationally, while it is more widely available in the US. Mefloquine's specific mechanism is not fully understood, but it is thought to interfere with protein synthesis of the *Plasmodium spp.* (80s subunit), and it may also interfere with hemoglobin digestion.

Primaquine is less common for prophylaxis than the other medications mentioned above. The primary mechanism disrupts mitochondria and binds to DNA within the *Plasm* ganism making it effective against exoerythrocytic stages of the protozoa.

CLINICAL CORRELATION

Most cases of malaria, as you have learned, are caused by *P. falciparum.* Primaquine use, for prophylaxis and treatment, is limited because it has activity primarily against *P. ovale and P. vivax.*

The best/ most common use of primaquine is for preventing relapses of *P. ovale and P. vivax.*

Anti-malarial Comparison

Drug	Frequency	Initiation (time before first exposure)	Discontinuation (time after last exposure)
Chloroquine/ Hydroxychloroquine	Once weekly	1-2 weeks	4 weeks
Atovaquone-proguanil	Once daily	1-2 days	7 days
Doxycycline	Once daily	1-2 days	4 weeks
Mefloquine	Once weekly	3 weeks (preferable) 2 weeks (acceptable)	4 weeks
Primaquine	Once daily	1-2 days	7 days

What are the adverse drug reactions of the drugs used for malaria prophylaxis? (LO#2)

Chloroquine and Hydroxychloroquine are generally well tolerated. One of the more common adverse effects is pruritus, though this often resolves within 48-72 hours. Although rare, there are some concerns for EKG changes and cardiomyopathy associated with these drugs, so use caution in patients with significant cardiac medical history. Lastly, avoid this medication in patients with a G6PD deficiency, as it can cause hemolysis in these patients.

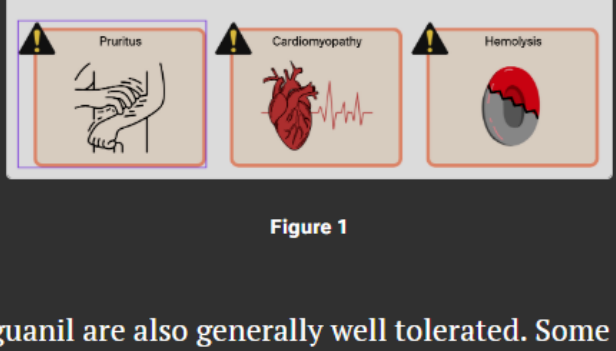
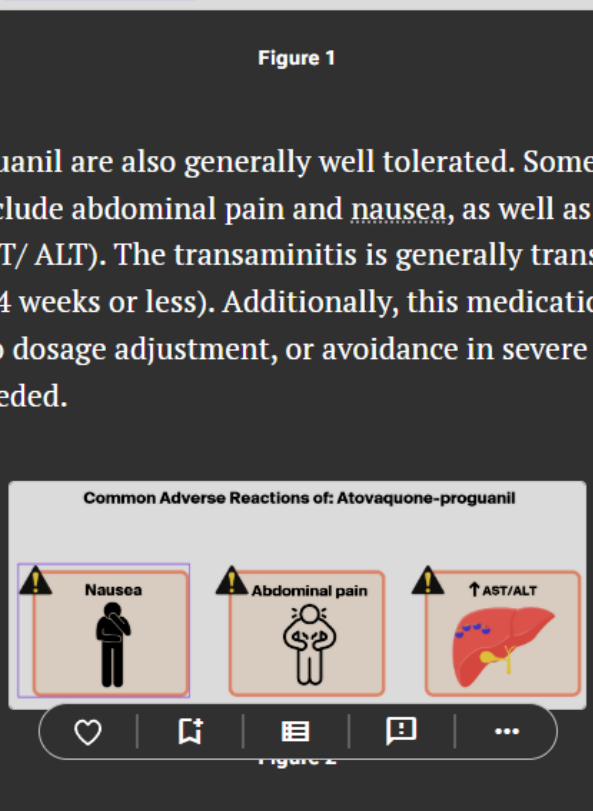

Common Adverse Reactions of: Chloroquine and Hydroxychloroquine
Pruritus
Cardiomyopathy
Hemolysis

Figure 1

Atovaquone-proguanil are also generally well tolerated. Some of the most common ADRs include abdominal pain and nausea, as well as increases in the liver enzymes (AST/ ALT). The transaminitis is generally transient and self-resolving (within 4 weeks or less). Additionally, this medication is eliminated via the kidneys, so dosage adjustment, or avoidance in severe renal impairment, is needed.


Common Adverse Reactions of: Atovaquone-proguanil
Nausea
Abdominal pain
↑AST/ALT

Doxycycline, as was previously discussed, is tolerated well with the exception of some potential GI disturbances (nausea, vomiting, diarrhea), photosensitivity, and contraindication when given with cations (causes chelation and makes it less effective).

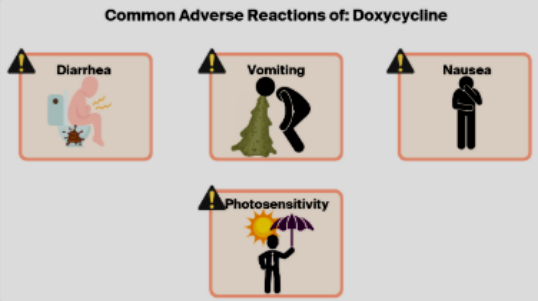


Figure 3

Mefloquine has some unique side effects that must be considered. It carries a black box warning for neuropsychiatric disorders, and is contraindicated in those patients with major psychiatric disorders, this includes a history of seizures. Additionally, mefloquine causes abnormal dreams and/ or insomnia. Often, these adverse effects limit the use of this medication.

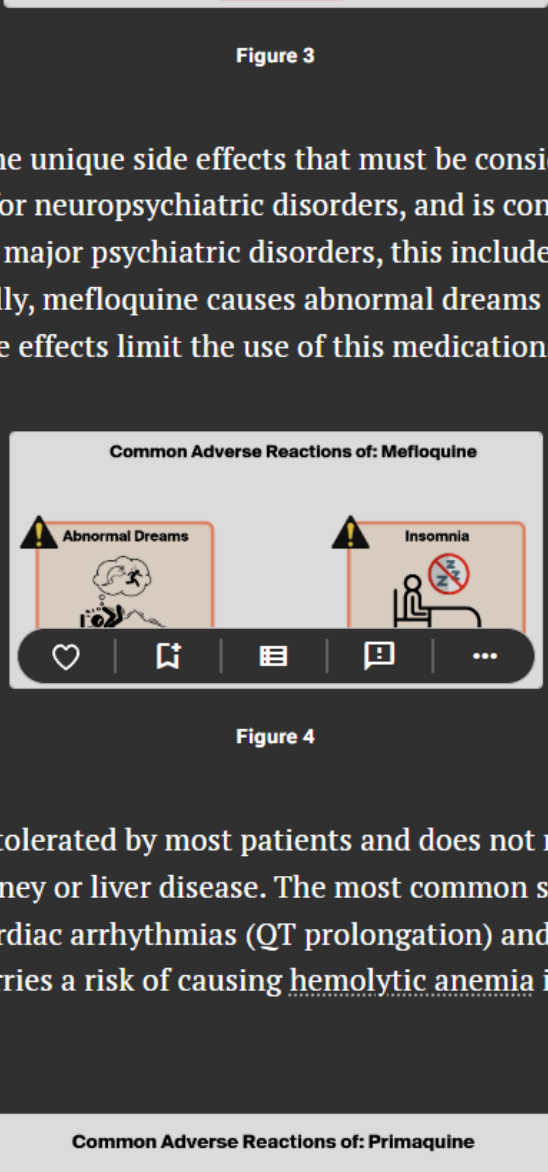

Common Adverse Reactions of: Mefloquine
Abnormal Dreams
Insomnia

Figure 4

Primaquine is well tolerated by most patients and does not require any dosage adjustments for kidney or liver disease. The most common side effects to be aware of include cardiac arrhythmias (QT prolongation) and leukopenia. Primaquine also carries a risk of causing hemolytic anemia in patients with G6PD deficiency.

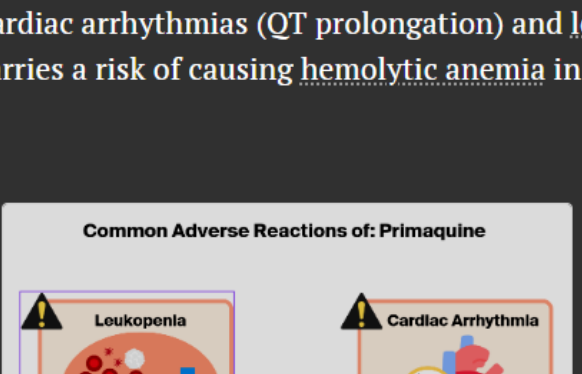

Common Adverse Reactions of: Primaquine
Leukopenia
Cardiac Arrhythmia

Figure 5

What is a major contraindication for mefloquine?

Malaria Treatment

How do drugs for malaria treatment work? (LO#1,3)

Uncomplicated Malaria. If a symptomatic infection occurs with parasitemia (<4% of RBCs infected) and no sign of organ dysfunction, the infection is considered uncomplicated. Antimalarial drug selection is then guided by whether infection was acquired in an area with chloroquine resistance. Chloroquine, or hydroxychloroquine, is still the preferred therapy in regions that do NOT have chloroquine resistance (e.g., Middle East). However, given the rate of resistance in most areas, first line therapy is recommended to be Artemisinin combination therapy (ACT). Most often these artemisinin medications are available in a single combination tablet; they are most often combined with the quinine derivatives lumefantrine, amodiaquine, or mefloquine.

Artemether (Oral only) and artesunate (Oral and IV availability) are the artemisinin medications that are readily available. Both of these medications work the same, however, as mentioned above they always come in combination tablets with other medications. Both formulations' mechanism is the same. They are metabolized to dihydroartemisinin (DHA). DHA is activated by heme iron binding, resulting in oxidative stress, inhibition of protein and nucleic acid synthesis, ultrastructural changes, and a decrease in growth and survival of the *Plasmodium spp*. The table on ACT therapies below highlights some of the most common ACT regimens available.

What is the first-line therapy for uncomplicated malaria in areas with chloroquine resistance?

ACT is preferred because they have a low adverse effect profile (discussed below), are effective (over 95% cure rate), and they have the most rapid clearance rate of the *Plasmodium spp.* relative to the other antimalarial drugs.

If ACT is not available, or there is a delay in acquiring ACT, the other medications mentioned above, atovaquone-proguanil, chloroquine/ hydroxychloroquine, doxycycline, mefloquine, and primaquine, are reasonable options, with consideration of resistance patterns based on patient exposure.

ACT Therapies

Medication	Route	Frequency/ Duration
Artemether-lumefantrine	Oral	Twice daily x 3 days
Artesunate-amodiaqui		Once daily x 3 days
Artesunate-mefloquine	Oral	Once daily x 3 days
Artesunate	IV	24 hours (then oral therapy)

In the absence of drug resistance, what is the preferred drug to treat uncomplicated malaria?

Complicated Malaria. Complicated malaria is characterized by hypoglycemia, severe anemia, thrombocytopenia, metabolic acidosis, and jaundice, with major signs of organ dysfunction (pulmonary edema, convulsions, etc) and >4% of RBCs infected. Intravenous (IV) medication is initiated for a minimum of 24 hours, and then the patient can take oral medication. IV artesunate is the preferred therapy for severe malaria. If IV artesunate is not readily available, fast-acting oral medications such as artemether-lumefantrine, atovaquone-proguanil, or mefloquine should be initiated immediately and continued until IV artesunate is available. IV or oral doxycycline should be avoided as it is slow-acting relative t

What are the adverse drug reactions of the drugs used for malaria treatment? (LO#2)

As was mentioned above, ACT has a low adverse effect profile, making it a preferred first line treatment option. The generalized adverse effects of hypotension, dizziness, and ataxia are the main ones to be on the lookout for.

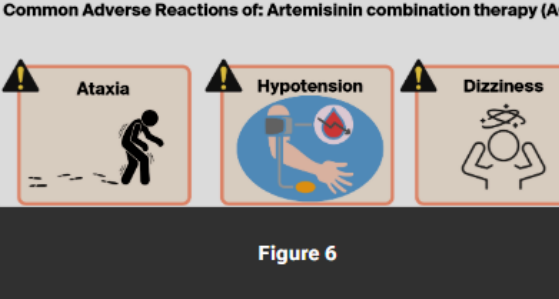

Common Adverse Reactions of: Artemisinin combination therapy (ACT)
Ataxia
Hypotension
Dizziness

Figure 6

Table 1 is a summary of facts for the most important drug therapies, including adverse effects and contraindications.

Table 1

Malaria Drugs Summary			
Drug and Its Uses	Mechanism of Action	Adverse Effects	Contraindications
Artemisinin	Cleaves Fe^{2+} of endoperoxide bridge and forms free radicals	Hypotension, dizziness, ataxia	Allergy to artemisinins
Atovaquone-proguanil	Inhibits parasite inhibits dihydrofolate reductase (proguanil)		Severe renal failure
Chloroquine	Inhibition of heme metabolism in parasite vacuoles, causing build-up of free heme	Pruritus, EKG changes/ cardiomyopathy, and hemolysis in G6PD deficiency	Allergy to chloroquine and related compounds
Doxycycline	Protein synthesis inhibitor (30s)	GI disturbances, photosensitivity	Allergy to tetracyclines and coadministration with cations (e.g., Ca++)
Mefloquine	Protein synthesis inhibitor (80s)	Abnormal dreams, insomnia	Allergy to similar compounds (mefloquine, quinine and quinidine), history of seizures or psychiatric history
Primaquine	Disrupts mitochondria and binds to DNA	Cardiac arrhythmia, leukopenia	G6PD deficiency (hemolytic anemia), patients at risk for granulocytopenia

How do we treat *Babesia spp.*? (LO#1,2,3)

As you have learned, *Babesia* infection (Babesiosis) is more common in immunocompromised individuals (e.g., patients with HIV). Though, it can occur in immunocompetent patients as well. Regardless of the patient's immune status, treatment of Babesiosis consists of the same medications, and they should be familiar!

The preferred treatment regimen consists of the macrolide antibiotic azithromycin AND atovaquone (discussed above). Remember that azithromycin is a protein synthesis ll tolerated, but some common adverse effects include QT prolongation and GI upset. Additionally, watching out for drug interactions (i.e., CYP3A4) is warranted.

An alternative regimen, if the above is not available/ tolerated includes clindamycin (again protein synthesis inhibitor at 50s subunit) AND mefloquine (discussed above). Adverse effects and interactions for clindamycin are similar to azithromycin discussed above.

How do we treat *Brucella spp.*? (LO#1,2,3)

Treatment of *Brucella* infection (Brucellosis) also includes a combination of some familiar drugs that we have talked about previously. The drugs used are doxycycline AND an aminoglycoside, usually gentamicin. Both of these medications are protein synthesis inhibitors at the 30s subunit. Tetracyclines, like doxycycline, are associated with GI upset and photosensitivity, as well as interactions with cations, like magnesium and iron, commonly found in multivitamins that can render the drug less effective due to chelation. Aminoglycosides have a number of adverse effects, most importantly nephrotoxicity and ototoxicity.

In the event that a patient cannot tolerate and aminoglycoside, a fluoroquinolone or sulfamethoxazole-trimethoprim is used with doxycycline instead.

A summary of treatments for babesiosis and brucellosis are included in table 2 below.

Organism	Treatment	Notes
Babesia spp.	First-line: Azithromycin + Atovaquone	Preferred regimen for most patients; usually well tolerated
	Alternative: Clindamycin + Mefloquine	Only used if first-line is contraindicated/ not tolerated
Brucella spp.	First-line: Doxycycline + Gentamicin	Preferred regimen
	Alternative: Doxycycline + Fluroquinolone or SMX-TMP	Only used if first-line is contraindicated/ not tolerated

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to BD, is traveling to Kenya—an area of East Africa where *Plasmodium falciparum* is known to have resistance to chloroquine—the most appropriate chemoprophylactic option would be atovaquone-proguanil. This medication is effective in regions with chloroquine-resistant malaria and is generally well tolerated. Atovaquone-proguanil should be started 1–2 days before travel to the malaria-endemic area, taken daily during the trip, and continued for 7 days after leaving the area to ensure protection against late liver-stage parasites. Other effective alternatives for this region include doxycycline or mefloquine, but atovaquone-proguanil is often preferred due to its shorter post-travel dosing period and lower risk of neuropsychiatric or gastrointestinal side effects.

Summary

- Malaria prophyla areas, with common drugs including atovaquone-proguanil, chloroquine/hydroxychloroquine, doxycycline, mefloquine, and primaquine.
- Chloroquine and hydroxychloroquine interfere with the Plasmodium spp. by increasing pH and affecting lysosomal degradation of hemoglobin, while hydroxychloroquine is also used for autoimmune disorders.
- Atovaquone-proguanil acts synergistically to inhibit mitochondrial electron transport and folate synthesis, affecting both liver and blood stages of malaria.
- Doxycycline inhibits protein synthesis at the 30s subunit, while mefloquine's mechanism is less understood but thought to interfere with protein synthesis and hemoglobin digestion.
- Primaquine disrupts mitochondria and binds to DNA, effective against exoerythrocytic stages of P. ovale and P. vivax, and is used to prevent relapses.
- Artemisinin combination therapy (ACT) is preferred for uncomplicated malaria due to its high efficacy and low adverse effect profile, while IV artesunate is preferred for complicated malaria.
- For *Babesia spp.* the first-line treatment is atovaquone plus azithromycin for most patients, with clindamycin plus quinine as an alternative.
- For *Brucella spp.* the first-line treatment is doxycycline plus an aminoglycoside, alternative therapy includes doxycycline and either a fluoroquinolone or sulfamethoxazole-trimethoprim.

Review Questions

1. Which drug acts by inhibiting mitochondrial electron transport in Plasmodium?

A. ● Atovaquone-proguanil
B. ○ Mefloquine
C. ○ Hydroxychloroquine
D. ○ Doxycycline
E. ○ Primaquine

Hide Explanation

The correct answer is Atovaquone-proguanil (A). Atovaquone selectively inhibits mitochondrial electron transport of the parasite. Mefloquine (B) interferes with protein synthesis. Hydroxychloroquine (C) increases pH in vacuoles. Doxycycline (D) inhibits protein synthesis at the 30s subunit. Primaquine (E) disrupts mitochondria and binds DNA.

2. Which drug for malaria prophylaxis is contraindicated in psychiatric disorders?

A. ● Mefloquine
B. ○ Atovaquone-proguanil
C. ○ Chloroquine
D. ○ Doxycycli
E. ○ Primaquine

Hide Explanation

The correct answer is mefloquine (A). Mefloquine is contraindicated due to risk of neuropsychiatric disorders. Atovaquone-proguanil (B) has gastrointestinal side effects but is generally well tolerated. Chloroquine (C) may cause hemolysis in G6PD deficiency. Doxycycline (D) causes photosensitivity. Primaquine (E) affects individuals with G6PD deficiency, causing hemolytic anemia.

3. Which drug causes QT prolongation and leukopenia?

A. ○ Hydroxychloroquine
B. ○ Mefloquine
C. ○ Artemisinin
D. ● Primaquine

Hide Explanation

The correct answer is Primaquine (D). Primaquine can cause cardiac arrhythmias, specifically QT prolongation, and leukopenia. Hydroxychloroquine (A) may cause pruritus and cardiom nia. Mefloquine (B) causes neuropsychiatric effects. Artemisinin (C) includes hypotension rather than QT prolongation.

4. Which medication is used with azithromycin for first-line Babesiosis treatment?

A. ● Atovaquone
B. ○ Doxycycline
C. ○ Metronidazole
D. ○ Clindamycin

Hide Explanation

The correct answer is atovaquone (A). Atovaquone is used in combination with azithromycin for treating Babesiosis. Doxycycline (B) is not part of the standard regimen for Babesiosis. Metronidazole (C) is used for anaerobic infections, not Babesiosis. Clindamycin (D) is not typically used for Babesiosis.

5. Which drug combination is used for Brucellosis treatment if aminoglycosides are intolerable?

A. ● Doxycycline + Ciprofloxacin
B. ○ Doxycycline
C. ○ Ciprofloxacin alone
D. ○ Sulfamethoxazole-Trimethoprim alone

Hide Explanation

The correct answer is doxycycline and ciprofloxacin (A). This combination is used when aminoglycosides are intolerable. Doxycycline and gentamicin (B) is the standard treatment. Fluoroquinolone alone (D) is not sufficient. Sulfamethoxazole-trimethoprim alone (E) lacks efficacy.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Justify the use of antimicrobial agents in treating *Plasmodium spp., Babesia spp., and Brucella spp.* based on their mechanism of action.~~

02 ☑ ~~Recogn used for *Plasmodium spp., Babesia spp., and Brucella spp.*~~

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Objective Confidence: 3/3